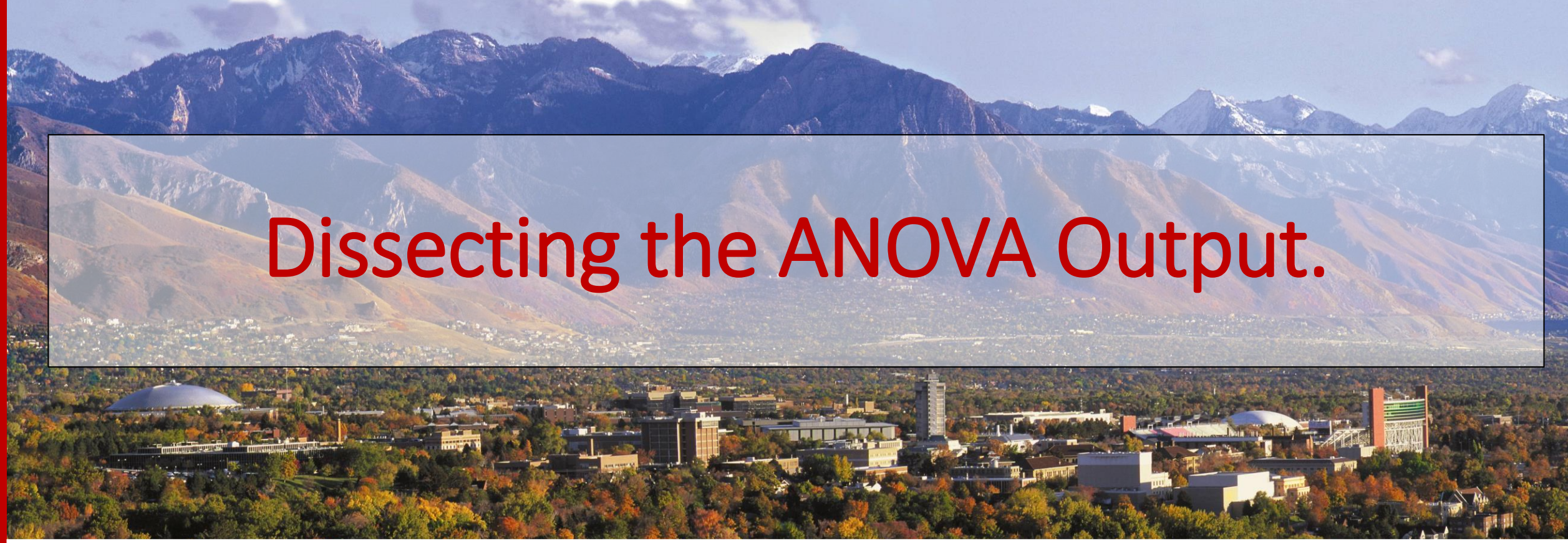

Dissecting the ANOVA Output.

Keith Lohse, PhD

Assistant Professor

Department of Health, Kinesiology, and Recreation

Department of Physical Therapy and Athletic Training

University of Utah

keith.lohse@health.utah.edu

@keith_lohse

HEALTH
UNIVERSITY OF UTAH

Comparing our Different Codes

Treatment or 'Dummy' Codes

```
Residuals:
   Min     1Q Median     3Q    Max
-7.964 -1.750 -0.500  2.250  8.286

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    68.500      1.779  38.497  < 2e-16 ***
groupsmedium   -6.750      2.516  -2.682 0.015209 *
groupsslower  -10.536      2.516  -4.187 0.000554 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.708 on 18 degrees of freedom
Multiple R-squared:  0.4999,    Adjusted R-squared:  0.4443
F-statistic: 8.996 on 2 and 18 DF,  p-value: 0.001957
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Contrast Codes

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MODEL A: $y_i = \beta_0 + \beta_1(X1_i) + \beta_2(X2_i) + \epsilon_i$

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Different models that make the same predictions:

<u>Treatment or 'Dummy' Codes</u>

$$68.500 - 6.750(X1_i) - 10.536(X2_i) =$$

<u>Contrast Codes</u>

$$62.7381 + 5.2679(X1_i) - 0.4940(X2_i) =$$

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$68.500 - 6.750(X1_i) - 10.536(X2_i) =$

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Note that although we estimate different coefficients depending on if X's are treatment coded or contrast coded, we arrive at the same model predictions.

This is because these coefficients are determined using least squares error estimation. Thus, the estimate we arrive at is the **mean** for each group.

The mean will produce the smallest SSE in each group and, as a result, the model will produce the smallest SSE overall.

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Despite different values of the coefficients, the SSE(A) for both of these models is the same and they are both being compared to the same SSE(C) to obtain the Omnibus F and PRE.

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… but the individual contrasts are different!

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Omnibus F and "post-hoc" tests.

- Consider that the false positive rate, α, is 0.05 (5%).

- If we wanted to test each of our groups compared to each other we have:
  - Slow v. Med, $\alpha = 0.05$
  - Med v. Fast, $\alpha = 0.05$
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  - So the "family-wise" false positive rate is now $3 * 0.05 = 0.15$ or 15%!

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  - So the "family-wise" false positive rate is now $3 * 0.05 = 0.15$ or 15%!

- In order to control the "family-wise" error rate, we will look at the omnibus F for a family of tests first.
  - If this omnibus F is statistically significant, then we will then interpret the individual contrast and conduct any additional **post-hoc tests**.